THE INTEGRAL OF THE SINGULAR PART OF THE JUMP MATRIX IS OF CUBIC ORDER

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Allowed standard prerequisites (editorial).

- The matching relation between the parametrices and the outer parametrix. same article, the display at source line 1377. Bounds the quotients appearing in the integrand and their inverses uniformly.
- The growth bounds of the local parametrices near the endpoint. same article, the display at source line 1381 and the decay estimate at source line 541. Provide the inverse square root bound and the exponential decay that together make the integrand small.
- The behaviour of the phase function at the endpoints. same article, the display at source line 429. Converts the distance to the endpoint into the variable in which the exponential decay is expressed.

Source reference. Original equation internal: sources/sigma-2024-004-recurrence-coefficients-for-orthogonal-polyn/pinned-original-source.tex, source lines 541–545. The counted statement or proof refers to that the author's displayed decay estimate for the exponential factor; it is the author's own earlier display, definition or statement in the same article and the whole argument of the counted span is retained verbatim.

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Source reference. Original equation internal: sources/sigma-2024-004-recurrence-coefficients-for-orthogonal-polyn/pinned-original-source.tex, source lines 429–433. The counted statement or proof refers to that the author's displayed behaviour of the phase function at the endpoints; it is the author's own earlier display, definition or statement in the same article and the whole argument of the counted span is retained verbatim.

Lemma 7.6. *For $n \to \infty$, we have*

$$\int_{\Sigma^U} \mu_n(v_n - \widehat{v}_n)\widehat{\mu}_n^{-1}\mathrm{d}s = O\left(\frac{1}{n^3}\right). \tag{7.1}$$

Proof. First, we need to show that $Q^\star(z) = Q(z)[P(z)]^{-1}$ for $z \in U$ is uniformly bounded as $n \to \infty$. Note that this statement is more intricate than the corresponding statements for $\widehat{Q}(z)\big[\widehat{P}(z)\big]^{-1}$ and $\widetilde{Q}(z)\big[\widetilde{P}(z)\big]^{-1}$, as the logarithmic weight function is not covered in [1] and hence we do not have equation (internal) for $Q(z)[P(z)]^{-1}$ at our disposal. Recall that for $s \in \Sigma^\ell = \partial U$, we have

$$\mu^\star(s) = Q^\star_-(s) = Q(s)\big[P(s)\big]^{-1}. \tag{7.2}$$

From (7.2), we see that on ∂U, $\mu^\star$ is in fact the restriction of the analytic function $Q^\star(z) = Q(z)[P(z)]^{-1}$, $z \in U$. Moreover, $\|\mu^\star\|_{L^2(\Sigma^U)} = O(1)$ for $n \to \infty$, as $\mu^\star = (1 - \mathcal{C}_{v^\star})^{-1}I$ and $(1 - \mathcal{C}_{v^\star})^{-1}$ is uniformly bounded on $L^2(\Sigma^\star)$. Thus, using Cauchy's integral formula, we get for $z \in U$

$$|Q^\star(z)| = \left|\frac{1}{2\pi\mathrm{i}}\int_{\partial U}\frac{\mu^\star(s)\mathrm{d}s}{s - z}\right| \lesssim O\left(\frac{1}{\operatorname{dist}(z, \partial U)}\right).$$

We have some freedom to choose U, so if necessary we can shrink it and conclude

$$|Q^\star(z)| = O(1), \qquad z \in U, \tag{7.3}$$

uniformly as $n \to \infty$. Let us now rewrite (7.1), where we drop the n-dependence for better readability:

$$\int_{\Sigma^U} \mu(v - \widehat{v})\widehat{\mu}^{-1}\mathrm{d}s = \int_{\Sigma^U} Q^\star(s)P_-(s)(v(s) - \widehat{v}(s))\big[\widehat{P}_-(s)\big]^{-1}\big[\widehat{Q}^\star(s)\big]^{-1}\mathrm{d}s. \tag{7.4}$$

We now list bounds for each of the factors in the integrand:

- (internal) and (internal) imply that for some $c > 0$

$$v(s) - \widehat{v}(s) = \begin{cases} \begin{pmatrix} 0 & 0 \\ \left(\frac{F^2}{w}(s) - \frac{\widehat{F}^2}{\widehat{w}}(s)\right)\phi^{-2n}(s) & 0 \end{pmatrix} \\ \quad = O\big(|s+1|^{3/2}\mathrm{e}^{-cn|s+1|^{1/2}}\big), & s \in \Sigma^U \setminus (-1, 1), \\ 0, \qquad s \in \Sigma^U \cap (-1, 1), \end{cases}$$

- (internal) and (7.3) implies that $[Q^\star(z)]^{\pm 1}, [\widehat{Q}^\star(z)]^{\pm 1} = O(1)$,
- and (internal) implies that $\big[P_-(z)\big]^{\pm 1}, \big[\widehat{P}_-(z)\big]^{\pm 1} = O\big(|z+1|^{-1/2}\big)$.

Taking the contributions of all factors in (7.4) into account we see that the integrand can be estimated by $O\big(|z+1|^{1/2}\mathrm{e}^{-cn|z+1|^{1/2}}\big)$, which precisely integrates to the error in (7.1). □

References

[1] Kuijlaars A.B.J., McLaughlin K.T.-R., Van Assche W., Vanlessen M., The Riemann–Hilbert approach to strong asymptotics for orthogonal polynomials on $[-1, 1]$, *Adv. Math.* **188** (2004), 337–398, arXiv:math.CA/0111252.